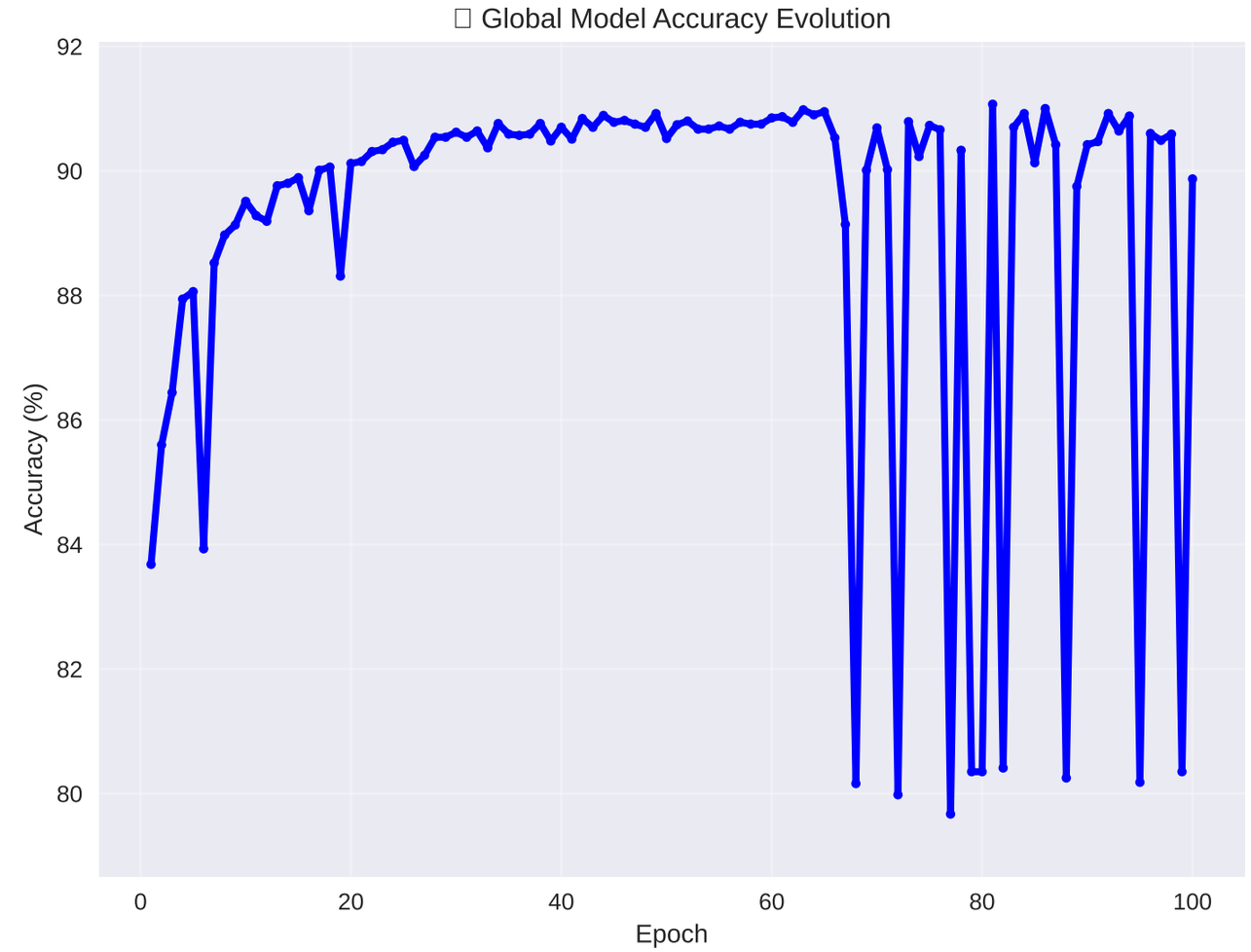
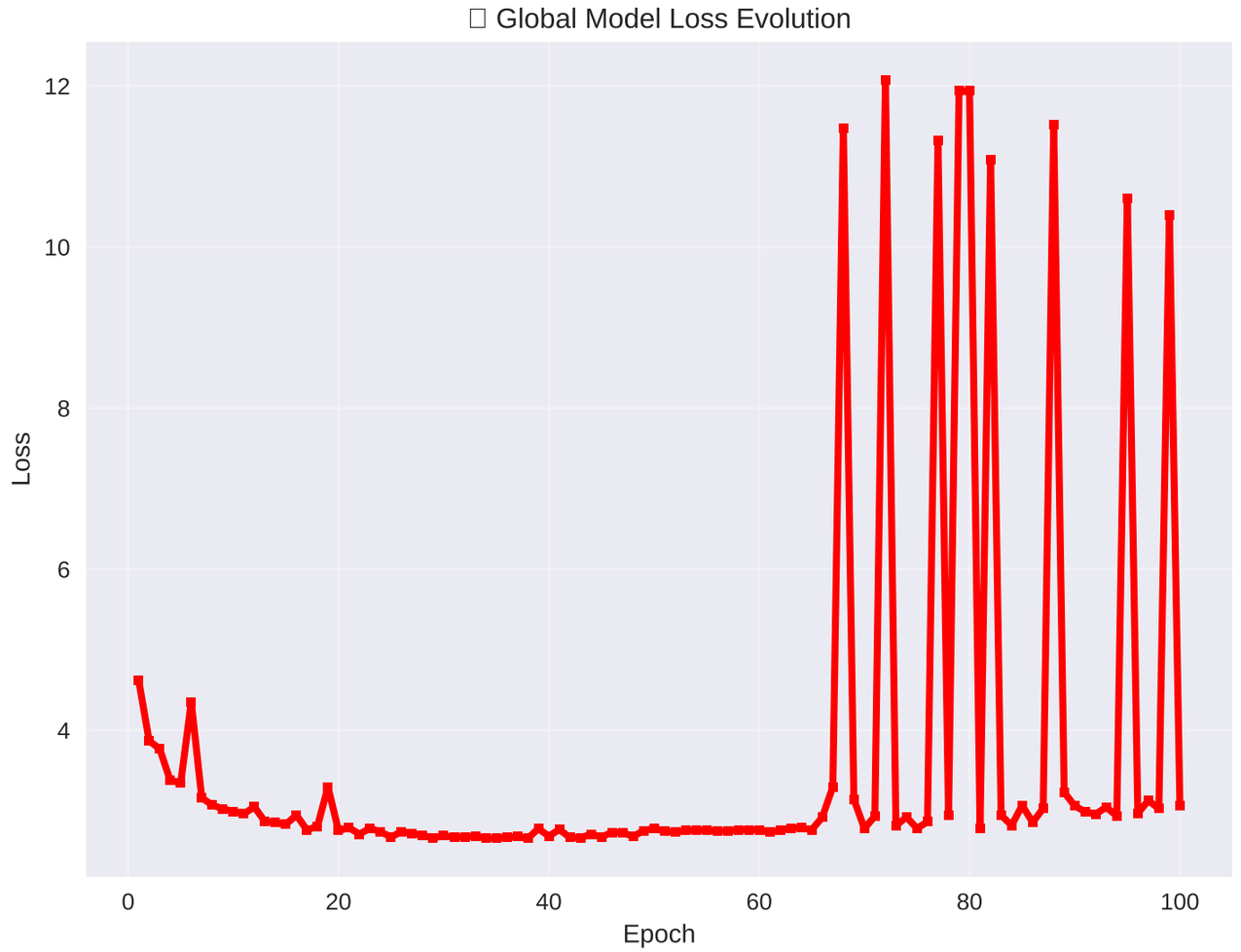
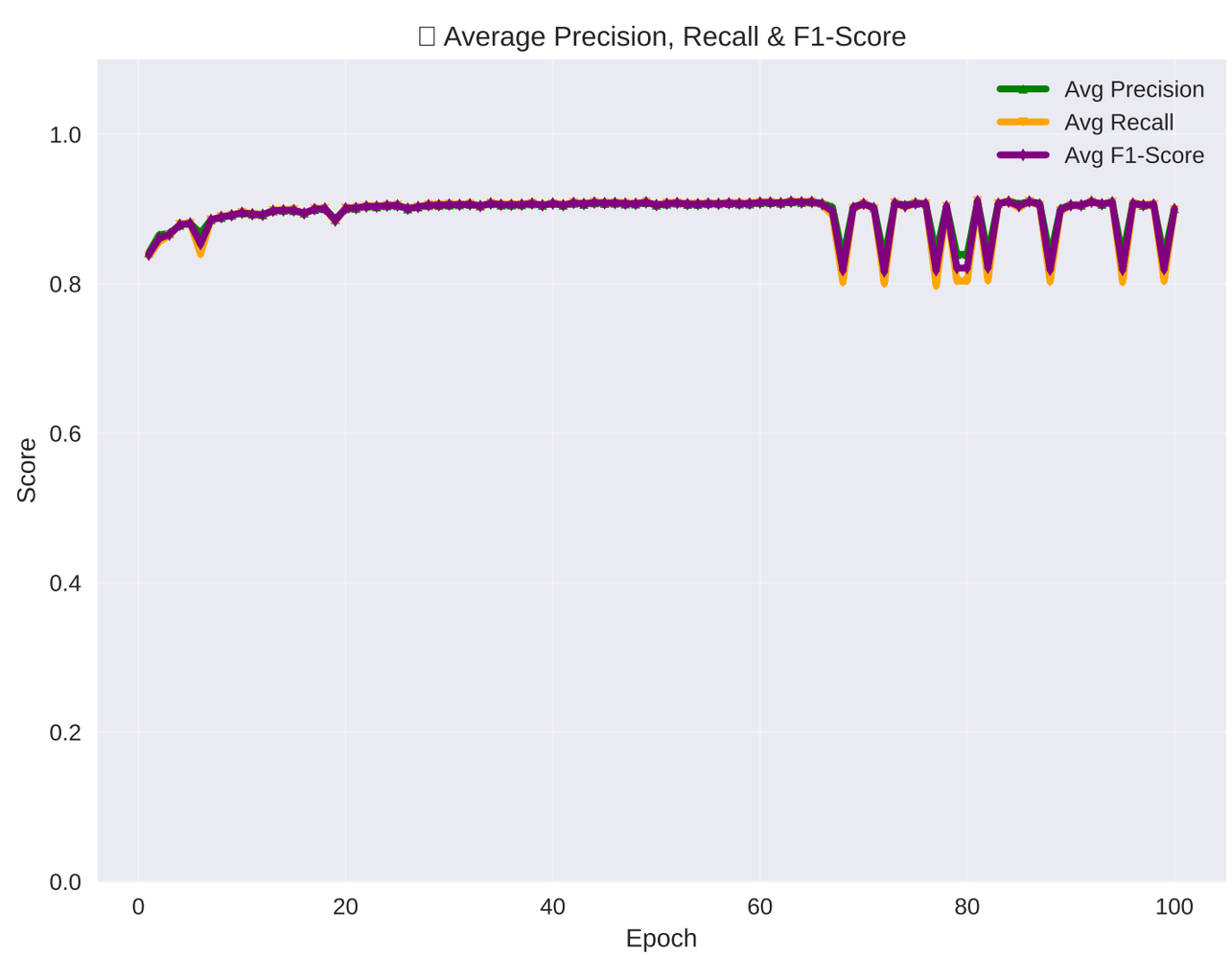
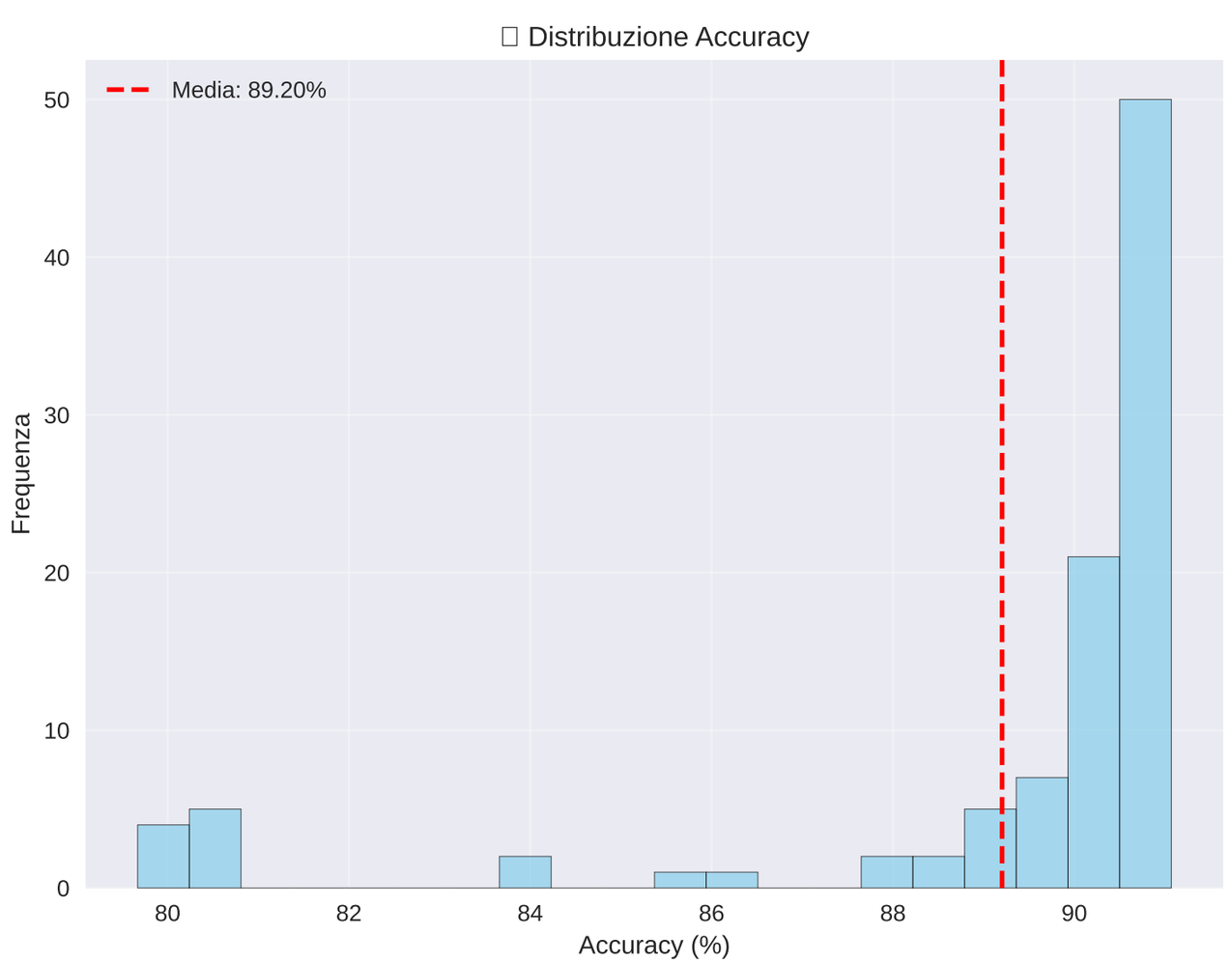
Federated Learning - Main Metrics Overview



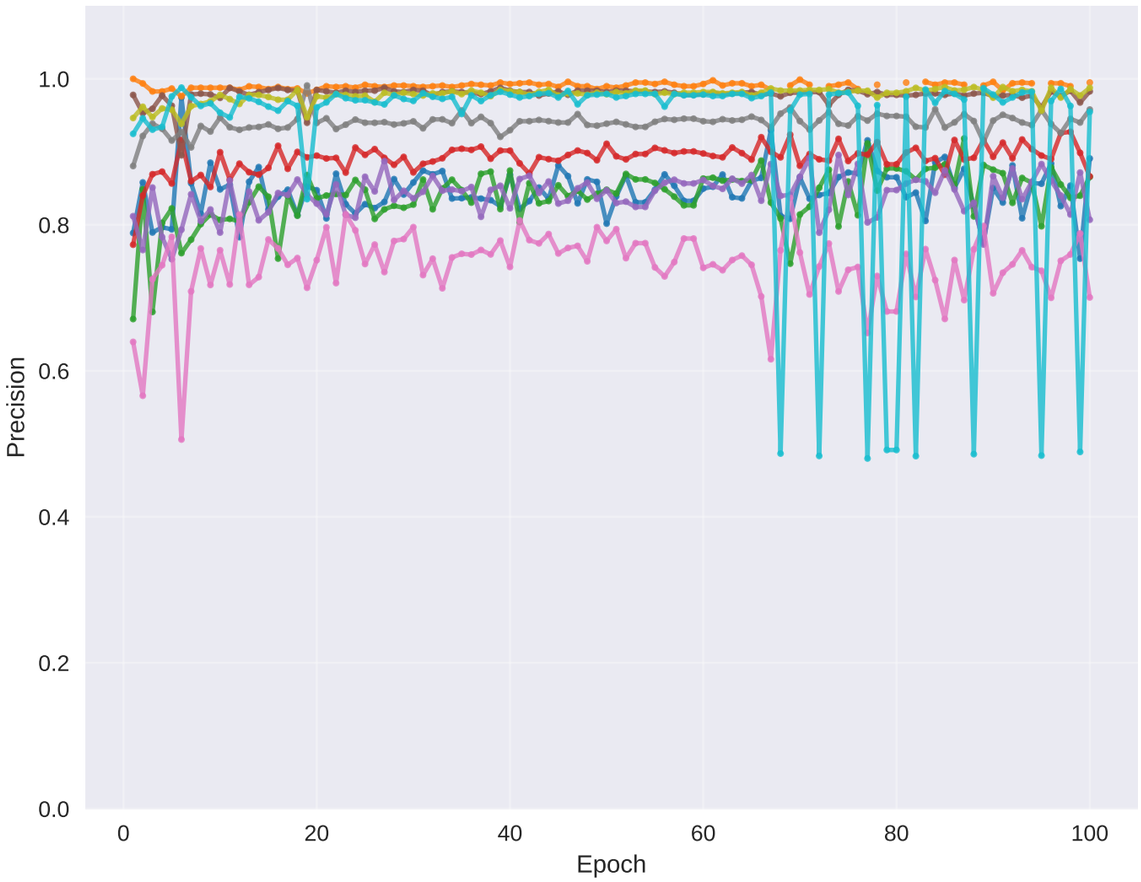




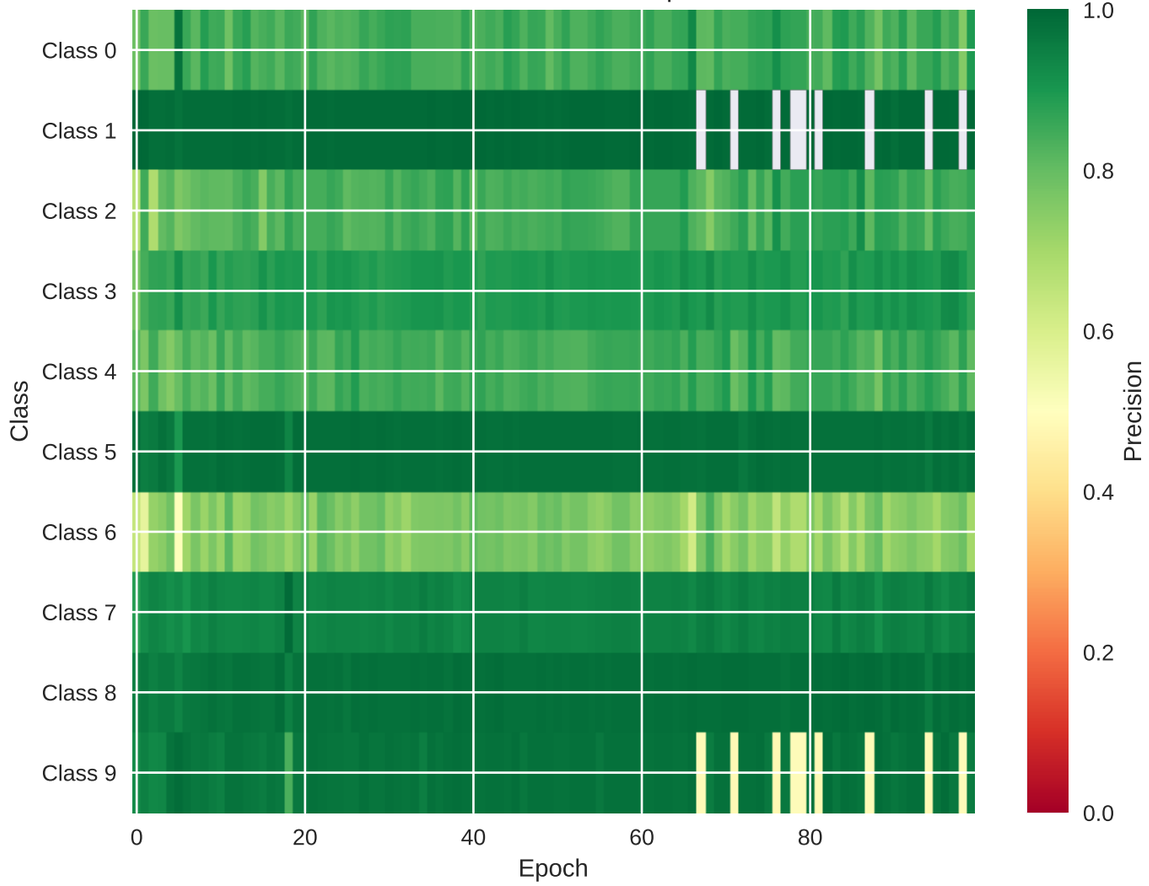


Per-Class Precision Analysis

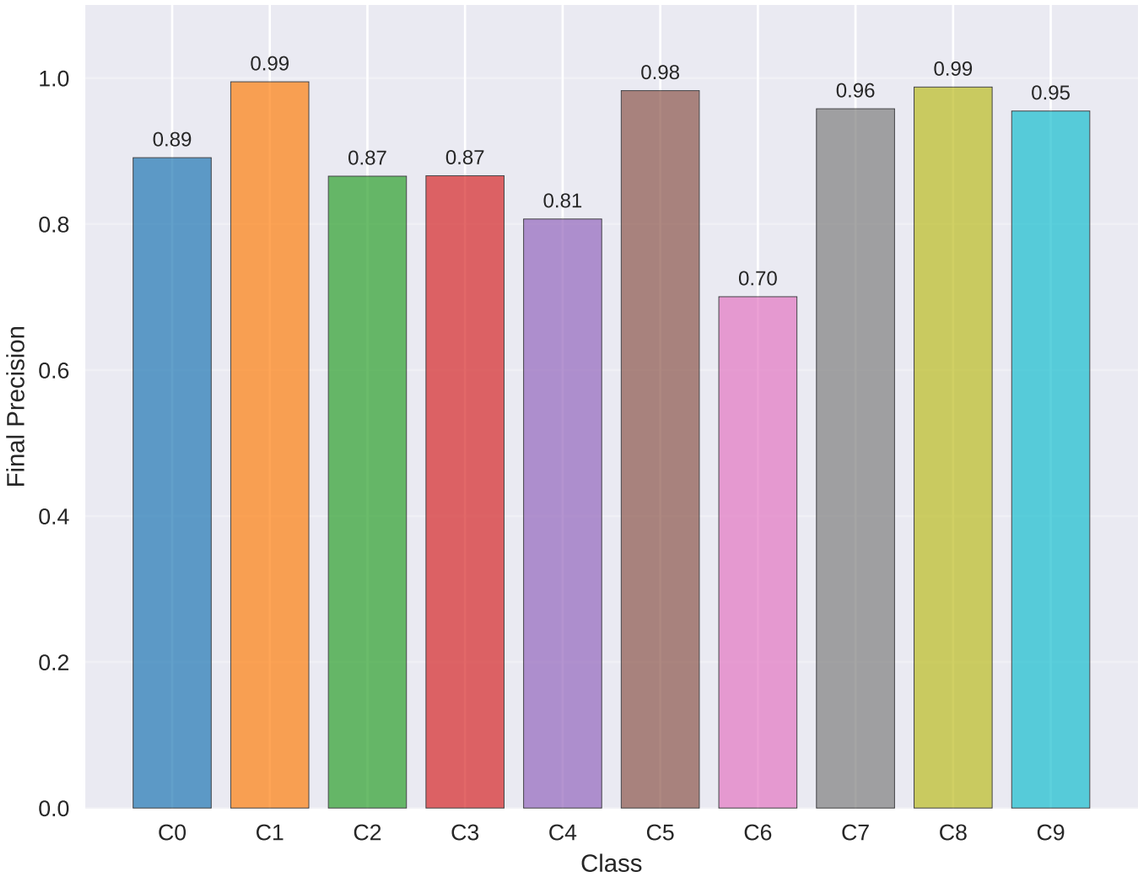
□ Per-Class Precision Evolution



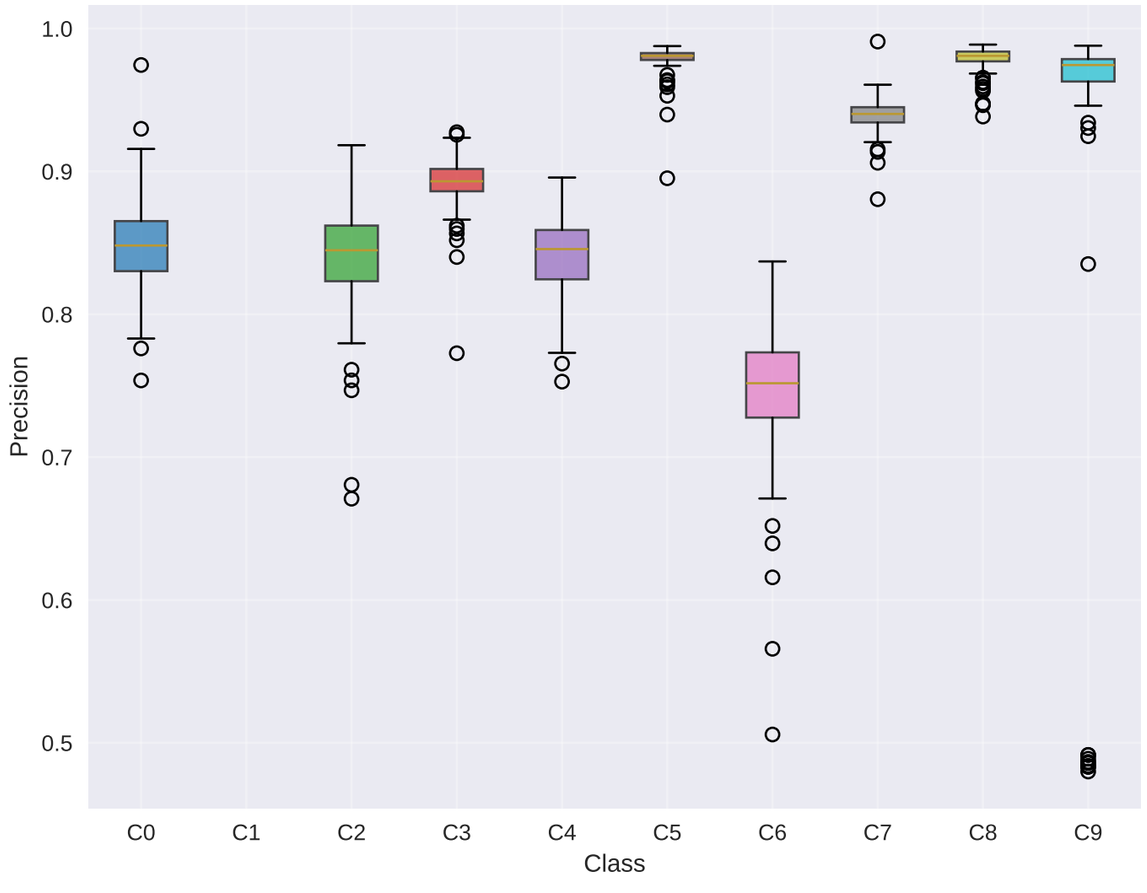
□ Precision Heatmap



□ Final Precision per Class

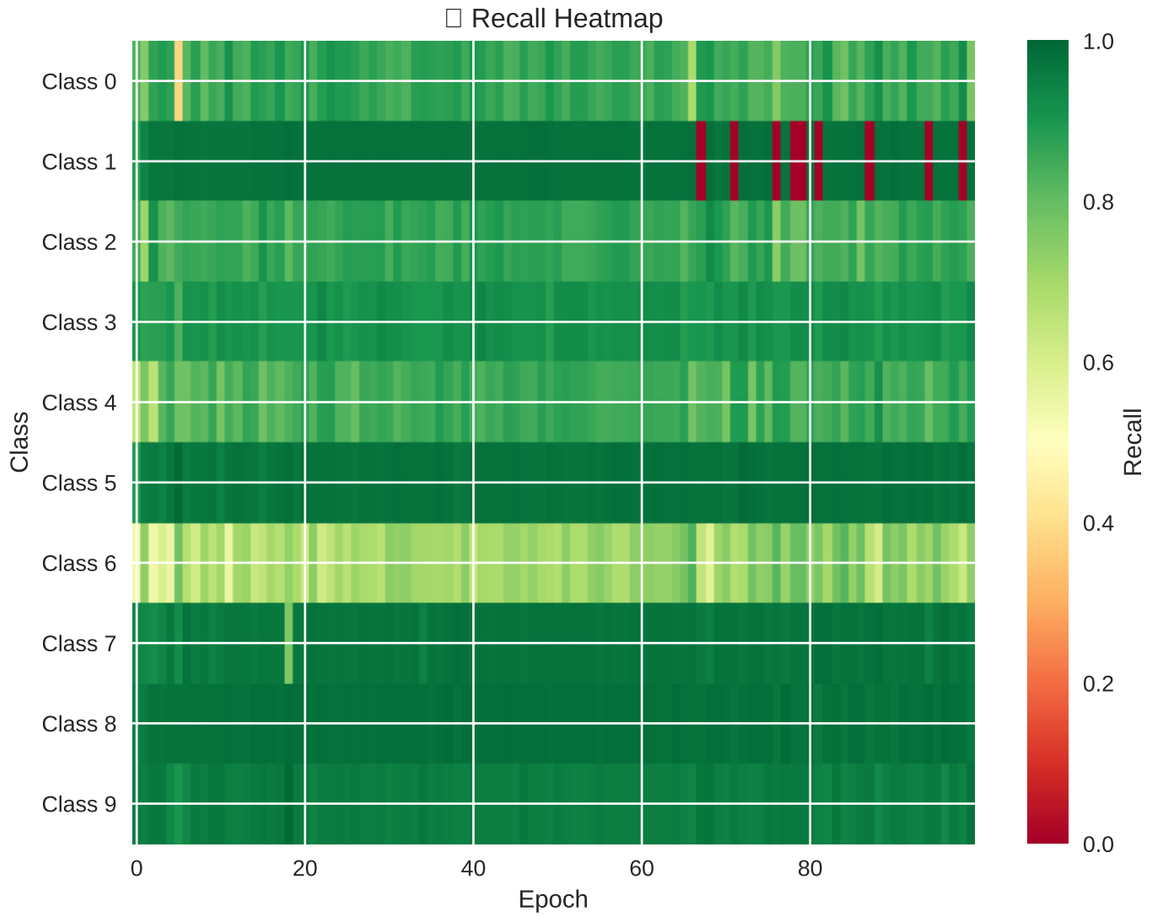


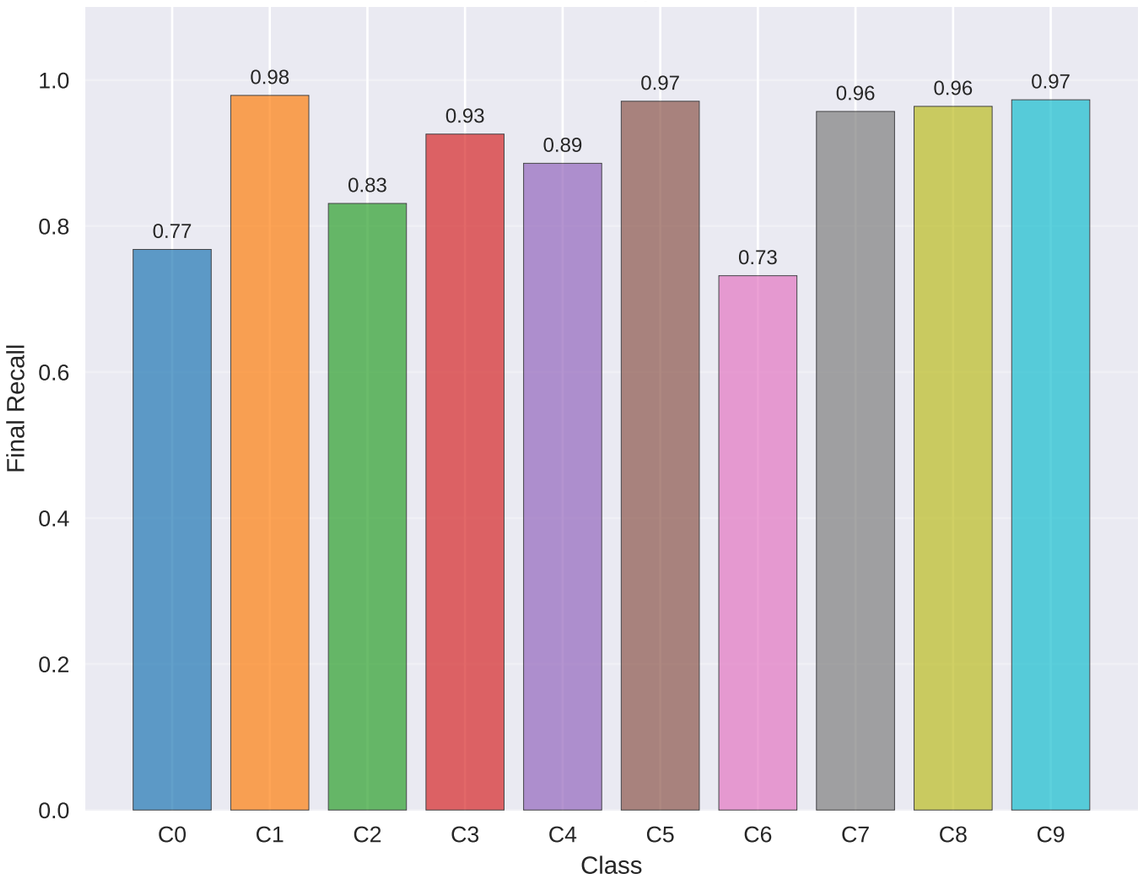
□ Precision Distribution per Class

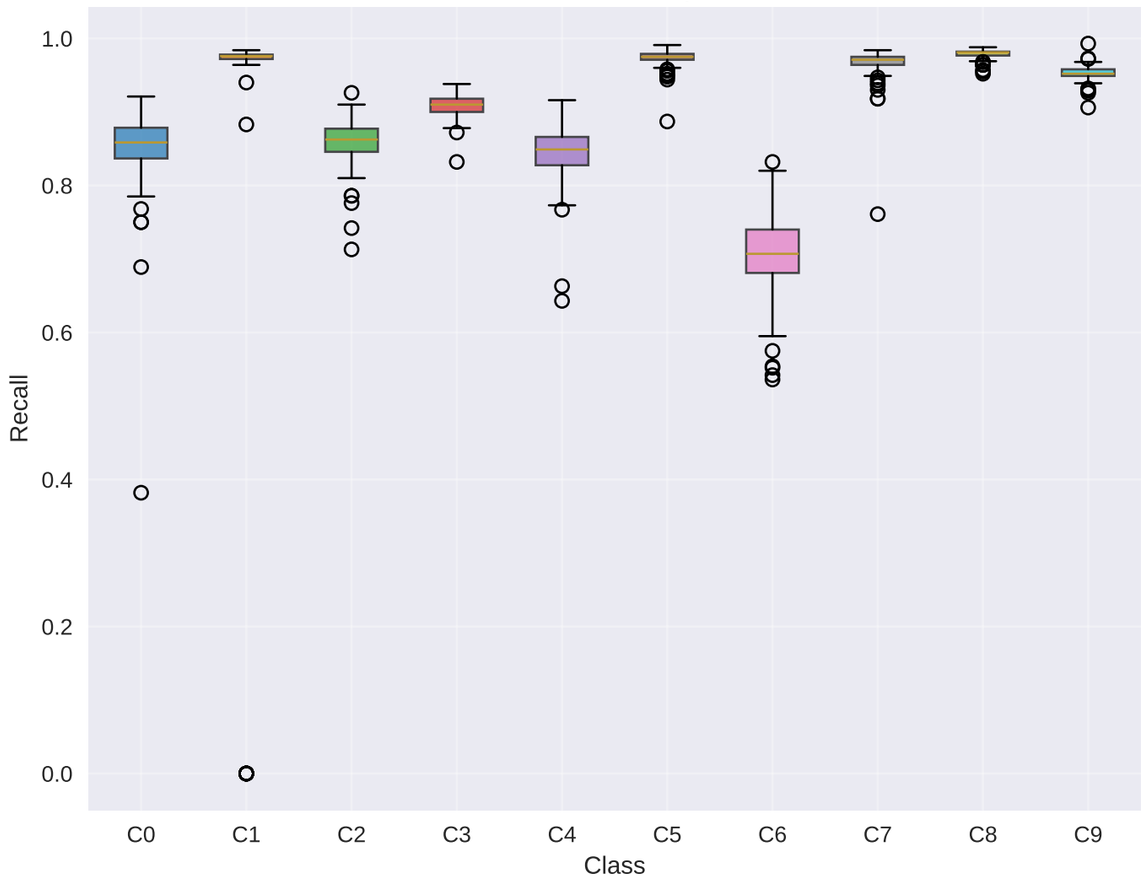


Per-Class Recall Analysis



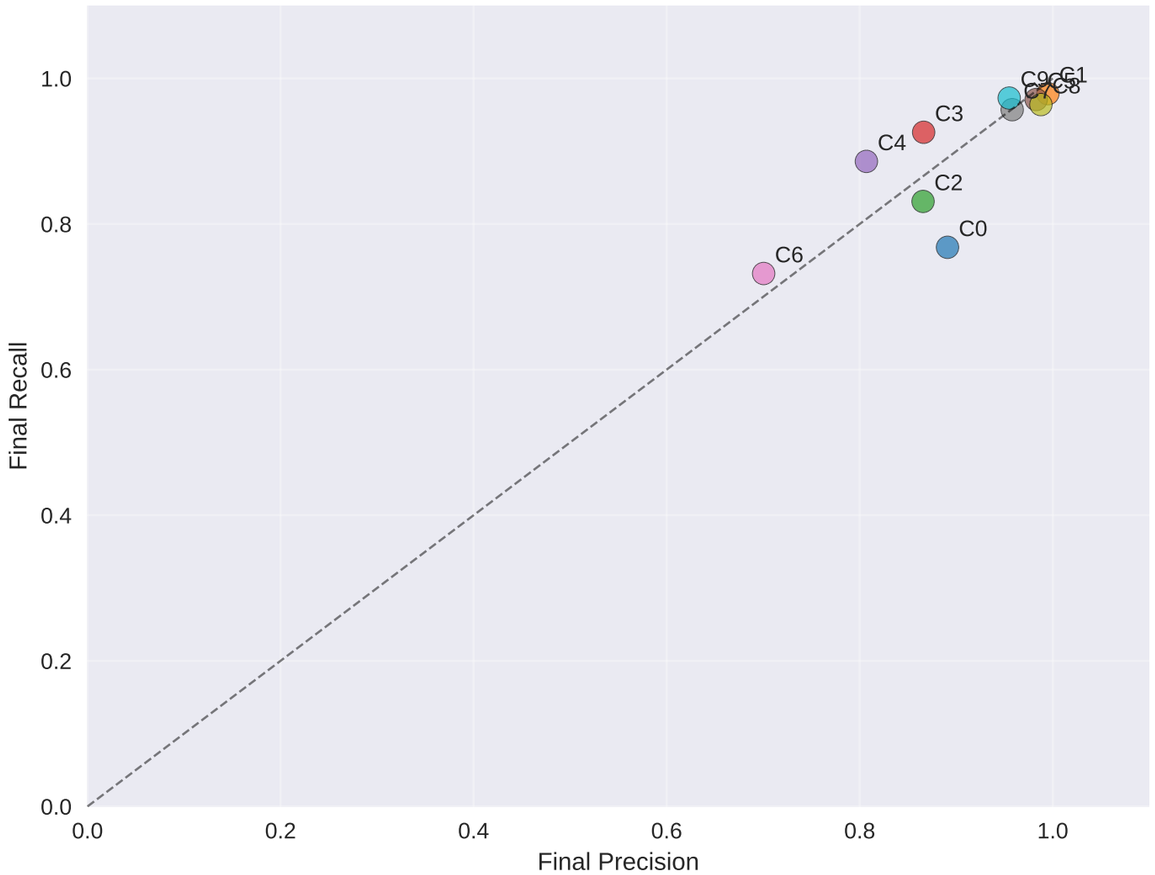




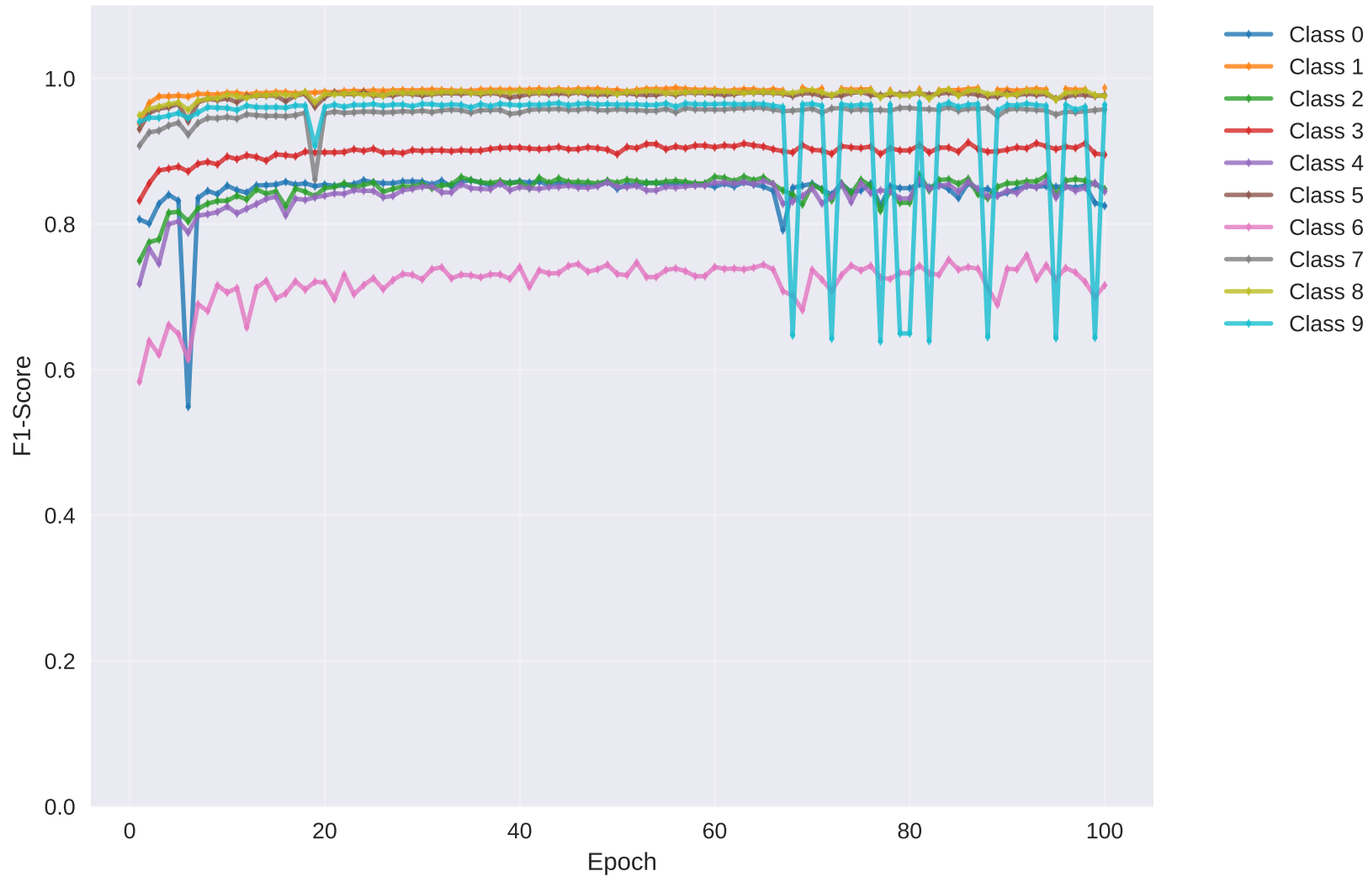


Comparative Analysis

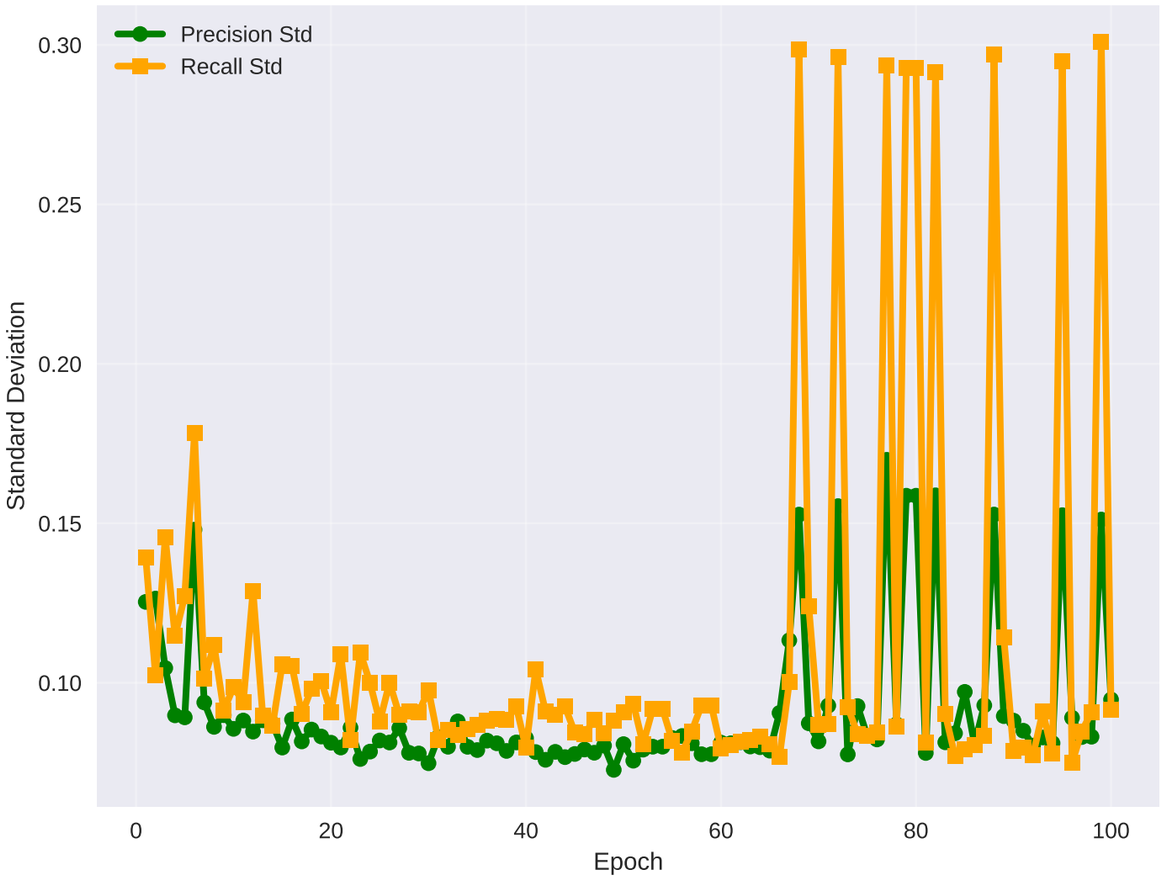
□ Precision vs Recall (Final Values)



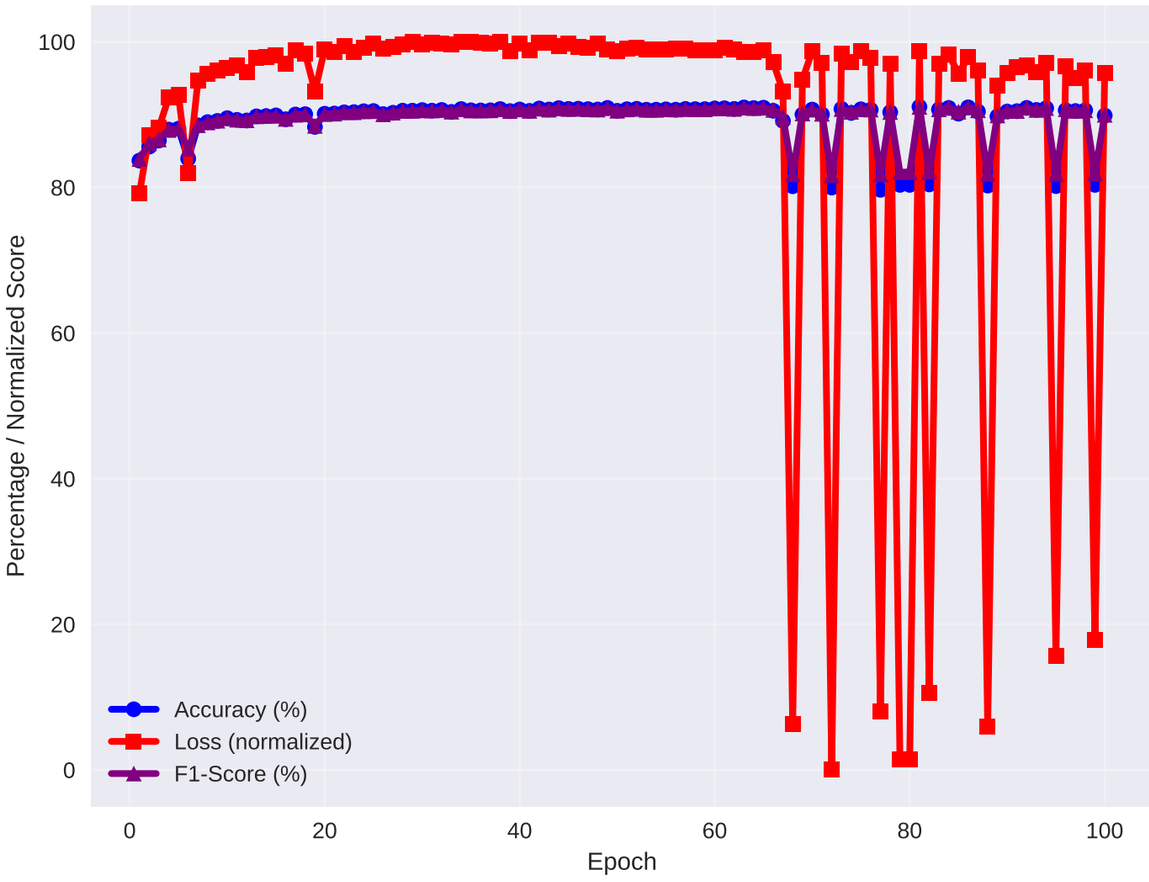
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 19:37:43
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 89.87%
• Loss: 3.0597
• Average Precision: 0.901
• Average Recall: 0.899
• Average F1-Score: 0.900

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.19%
• Riduzione Loss: +1.5619
• Stabilità Accuracy (CV): 0.0349

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.995)
• Peggiore Precision: Classe 6 (0.700)
• Migliore Recall: Classe 1 (0.979)
• Peggiore Recall: Classe 6 (0.732)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.891, R=0.768, F1=0.825
• Classe 1: P=0.995, R=0.979, F1=0.987
• Classe 2: P=0.866, R=0.831, F1=0.848
• Classe 3: P=0.866, R=0.926, F1=0.895
• Classe 4: P=0.807, R=0.886, F1=0.845
• Classe 5: P=0.983, R=0.971, F1=0.977
• Classe 6: P=0.700, R=0.732, F1=0.716
• Classe 7: P=0.958, R=0.957, F1=0.957
• Classe 8: P=0.988, R=0.964, F1=0.976
• Classe 9: P=0.955, R=0.973, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: MEDIA
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
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